Exercise 6 (b)

- Sensor Fusion (Attitude Heading and Reference System (AHRS)

- Sensor Fusion (Attitude Heading and Reference System (AHRS)

→ ▪ Inertial Measurement Unit (IMU) Sensor

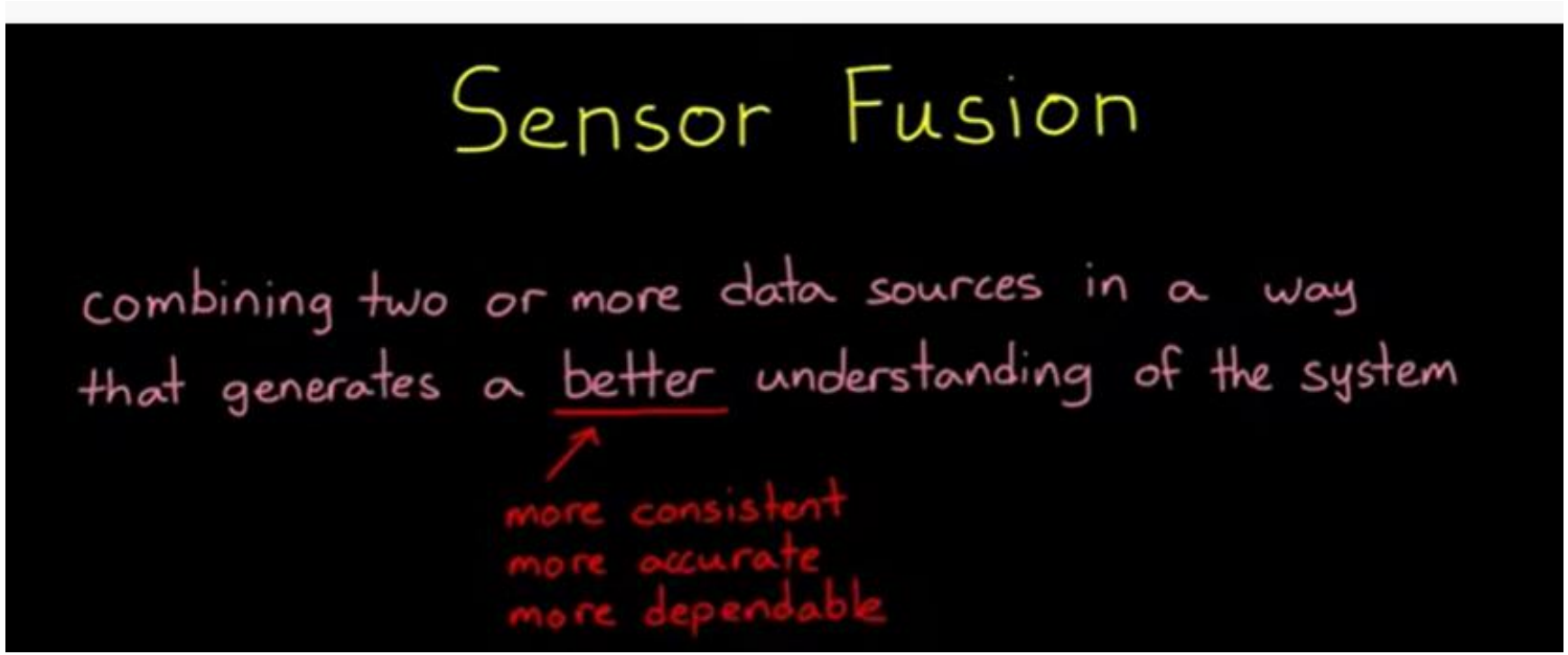
→ ▪ Orientation Estimation from IMU

▪ Sensor Fusion for Orientation Estimation

- Sensor Fusion (Attitude Heading and Reference System (AHRS)

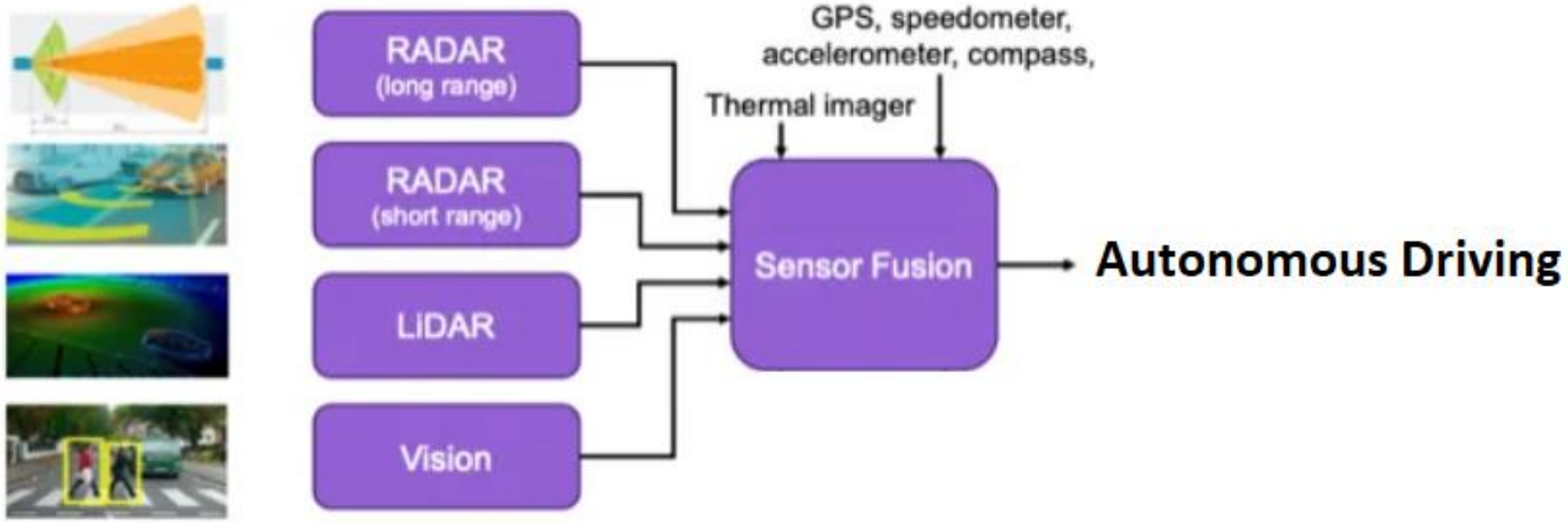
  - Inertial Measurement Unit (IMU) Sensor
  - Orientation Estimation from IMU
  - Sensor Fusion for Orientation Estimation

• Sensor Fusion (Attitude Heading and Reference System (AHRS)



Courtesy : Youtube Channel: *Matlab Tech Talk* with Brian Douglas

- Sensor Fusion (Attitude Heading and Reference System (AHRS)



• Sensor Fusion (Attitude Heading and Reference System (AHRS)

- Sensor Fusion to estimate orientation

• Sensor Fusion (Attitude Heading and Reference System (AHRS)

- Sensor Fusion to estimate orientation
  - **Attitude**

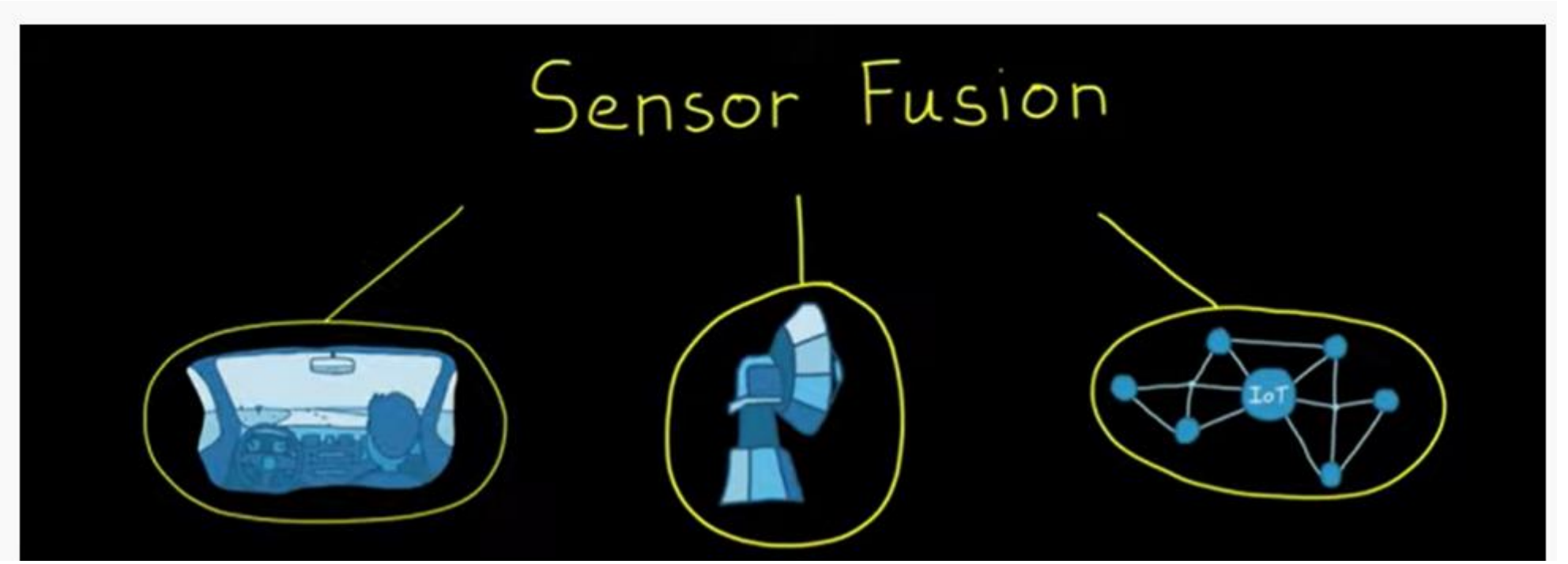
• Sensor Fusion (Attitude Heading and Reference System (AHRS)

- Sensor Fusion to estimate orientation
  - **Attitude**
  - **Heading**

• Sensor Fusion (Attitude Heading and Reference System (AHRS)

- Sensor Fusion to estimate orientation
    - Attitude Heading and Reference System (AHRS)

• Sensor Fusion (Attitude Heading and Reference System (AHRS)



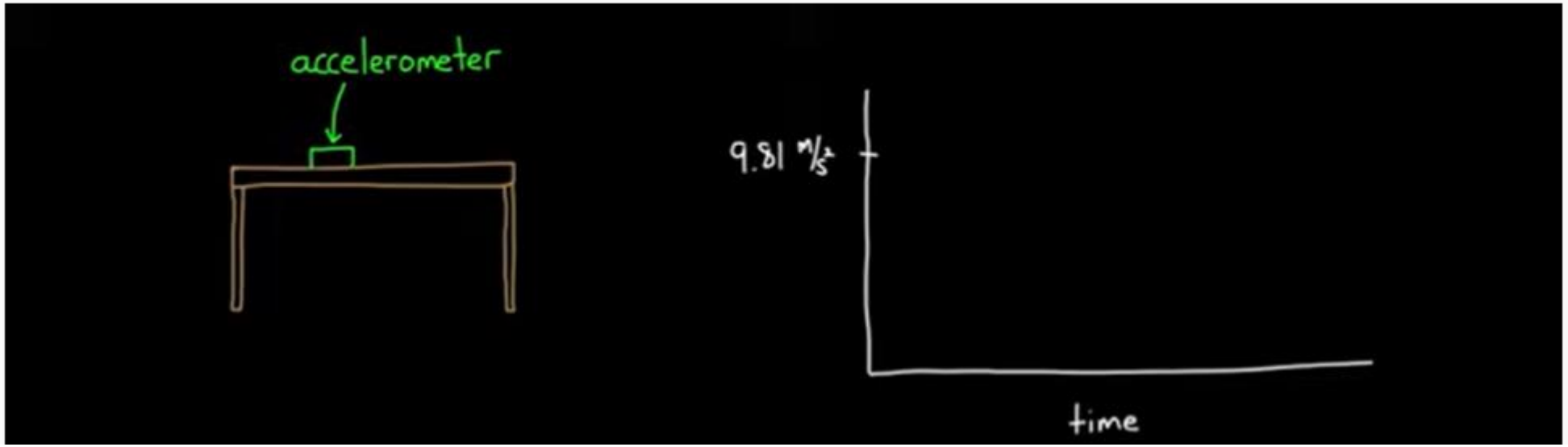
Courtesy : Youtube Channel: *Matlab Tech Talk* with Brian Douglas

- Sensor Fusion is an integeral part of any autonoumosu systems.
- All autonmous stystems rely on sensor fusion of one sort or another.

• Sensor Fusion (Attitude Heading and Reference System (AHRS)

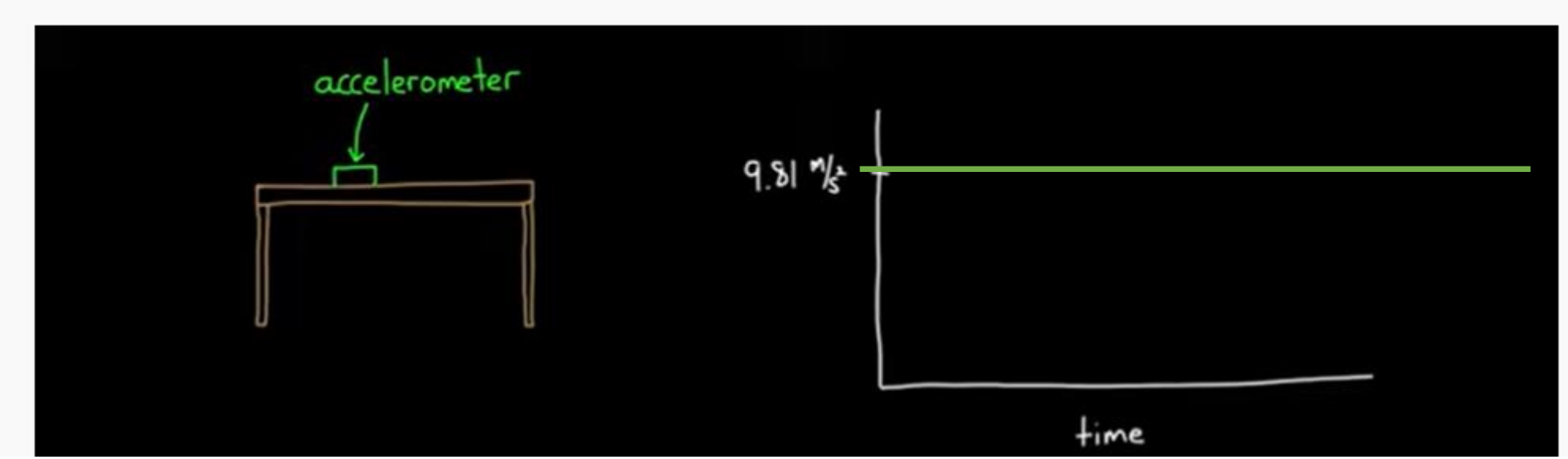
- To Increase the quality of the data.

• Sensor Fusion (Attitude Heading and Reference System (AHRS)



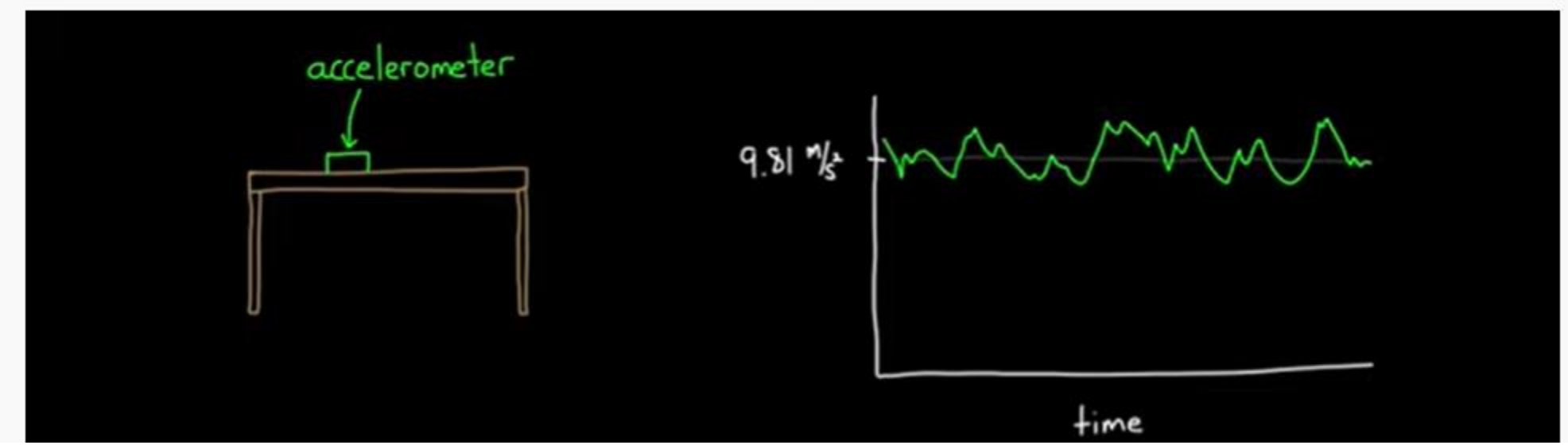
Courtesy : Youtube Channel: *Matlab Tech Talk* with Brian Douglas

• Sensor Fusion (Attitude Heading and Reference System (AHRS)



Courtesy : Youtube Channel: *Matlab Tech Talk* with Brian Douglas

• Sensor Fusion (Attitude Heading and Reference System (AHRS)



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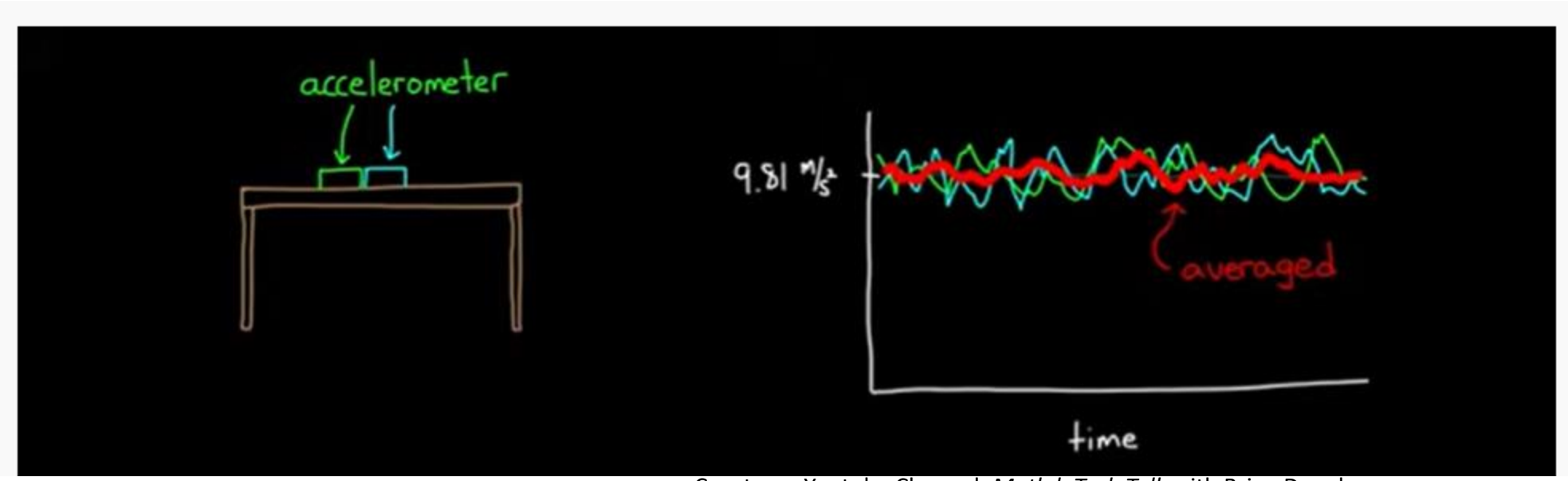

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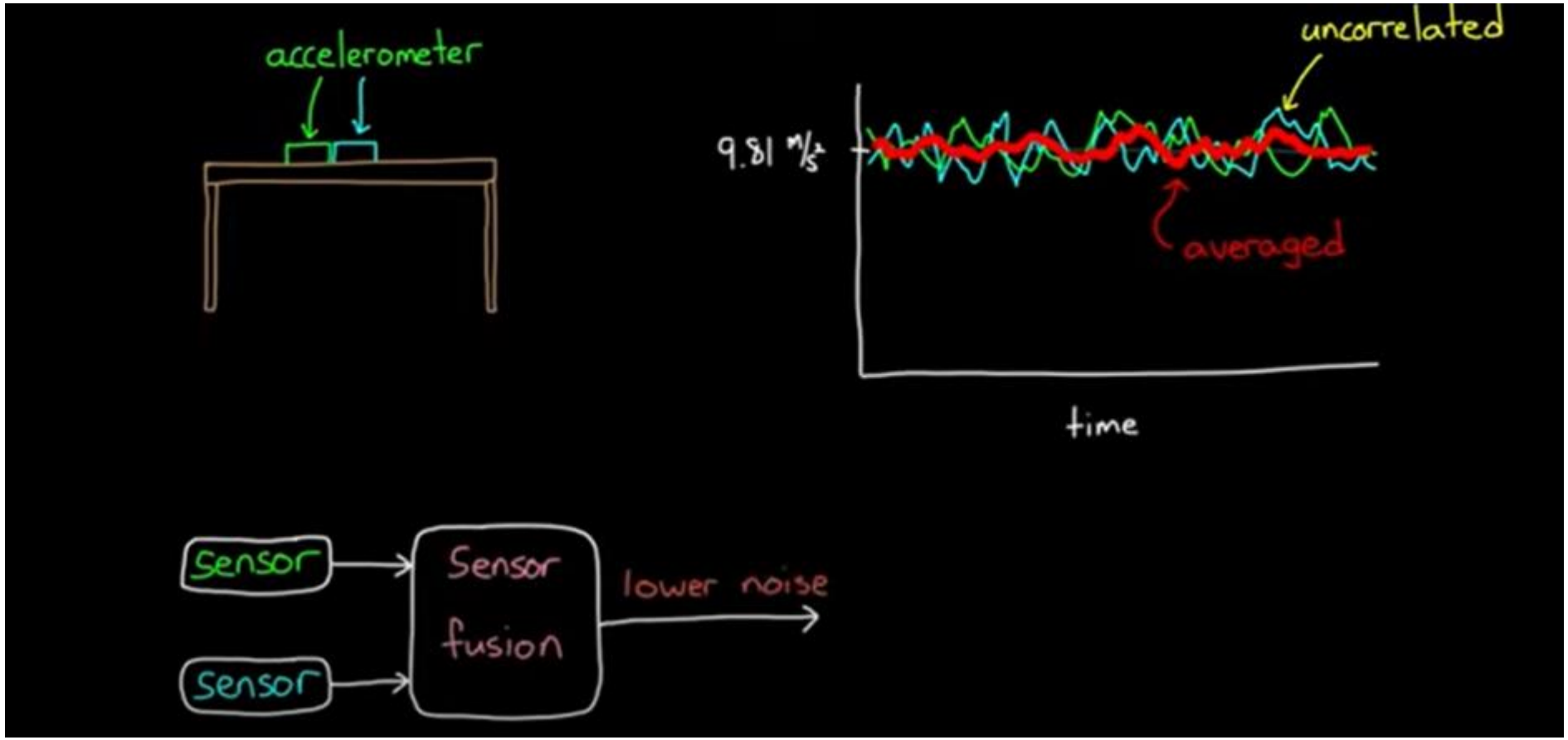
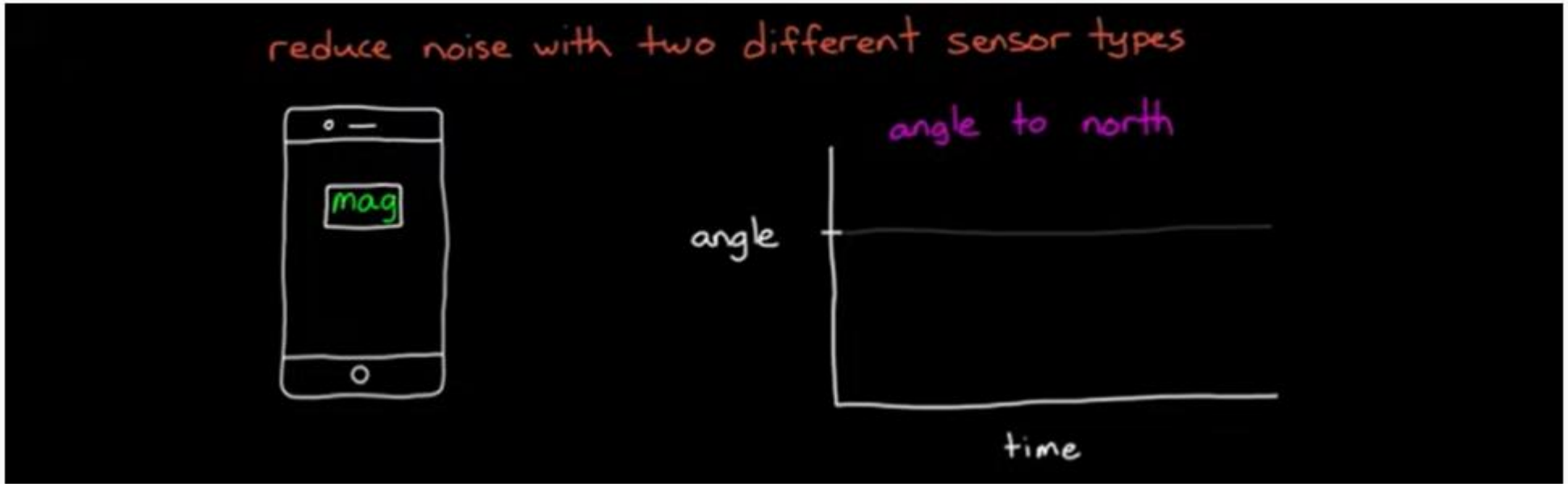


Courtesy : Youtube Channel: *Matlab Tech Talk* with Brian Douglas

• Sensor Fusion (Attitude Heading and Reference System (AHRS)

- Reducing the noise with two different types of sensors.



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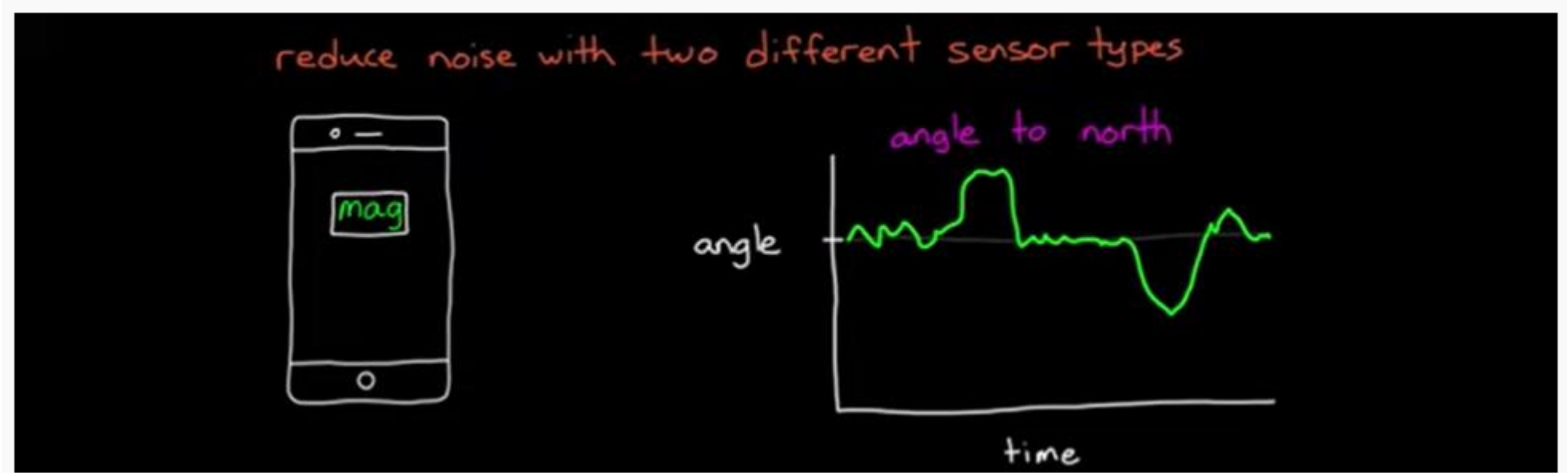
• Sensor Fusion (Attitude Heading and Reference System (AHRS)



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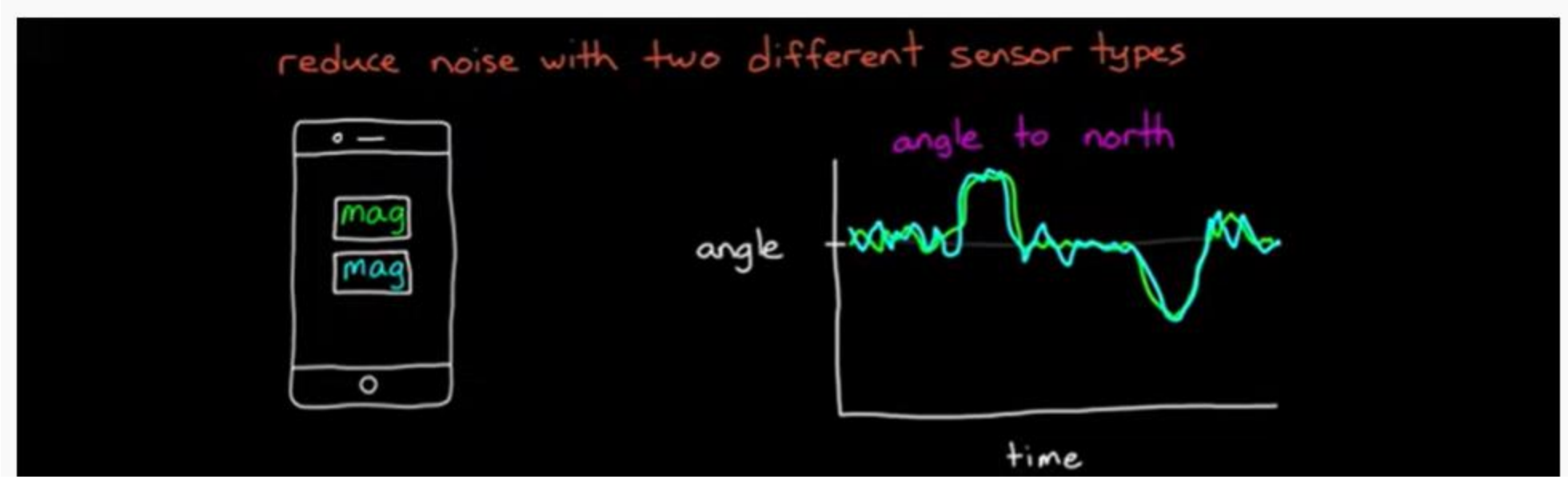

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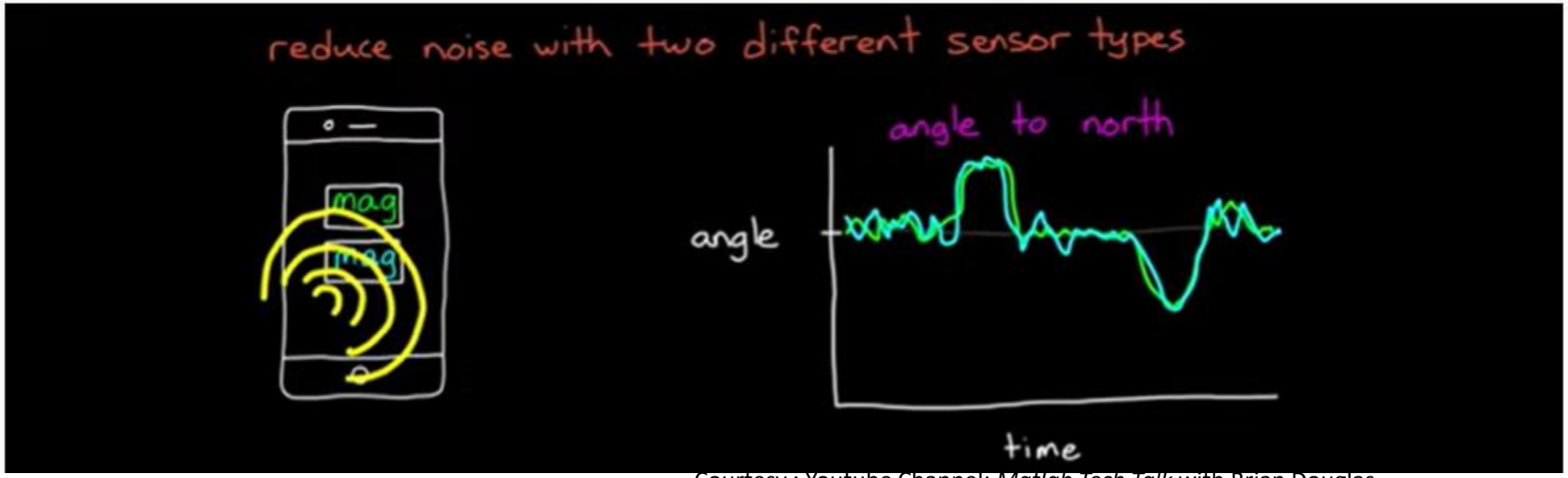

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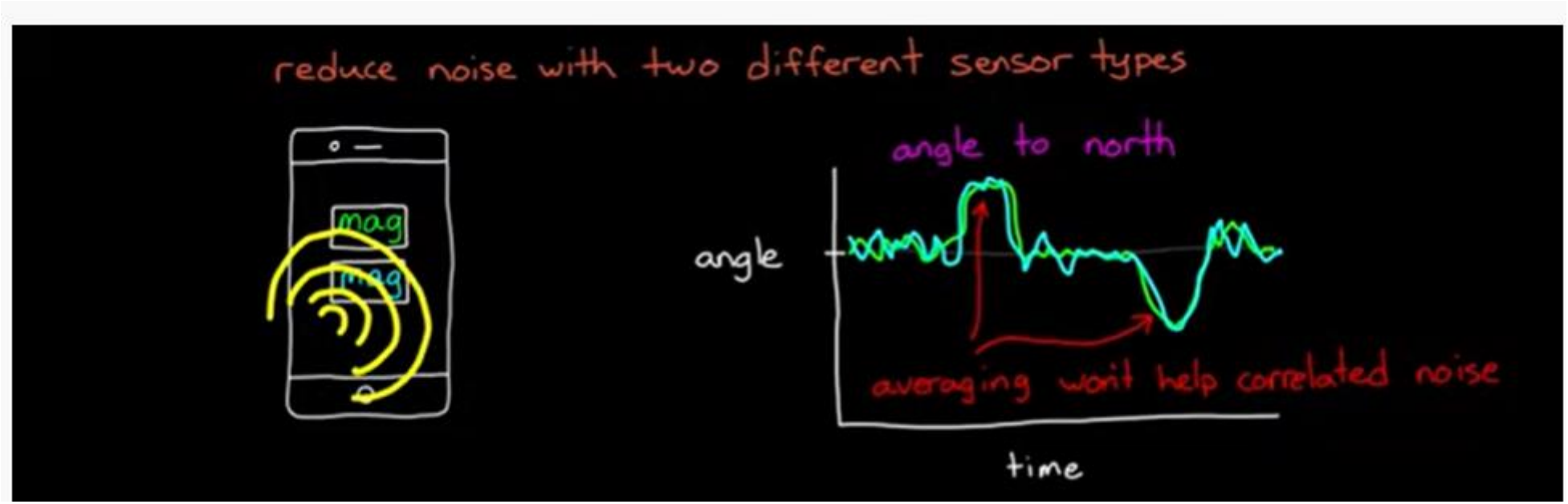

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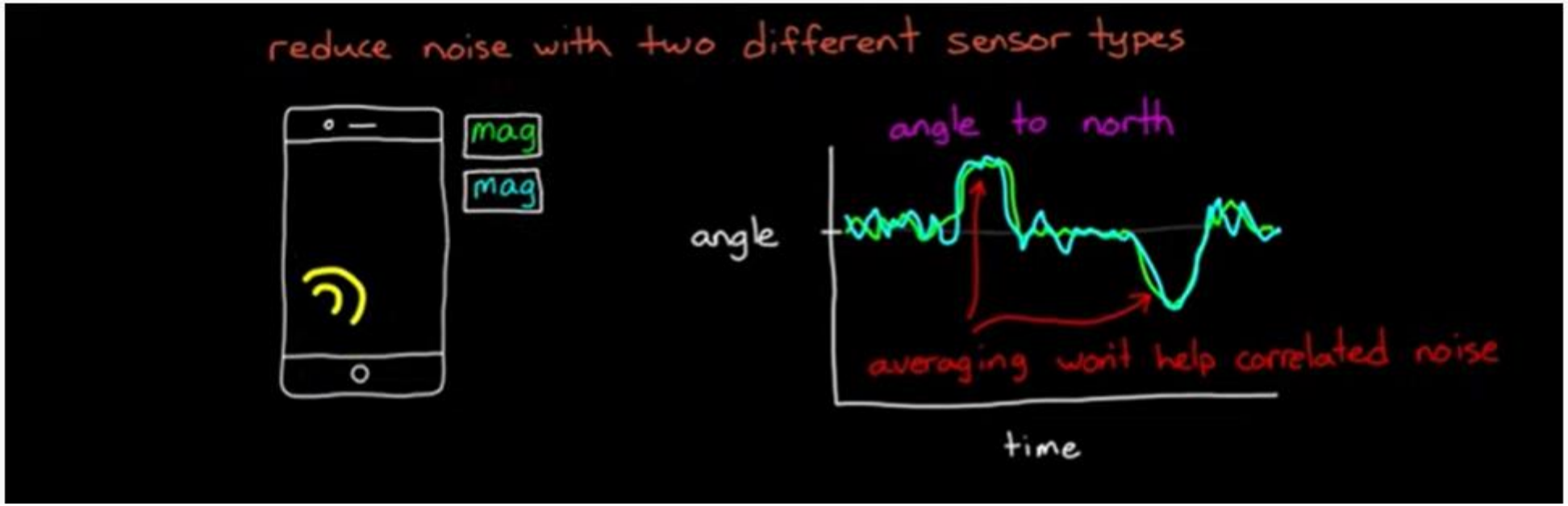

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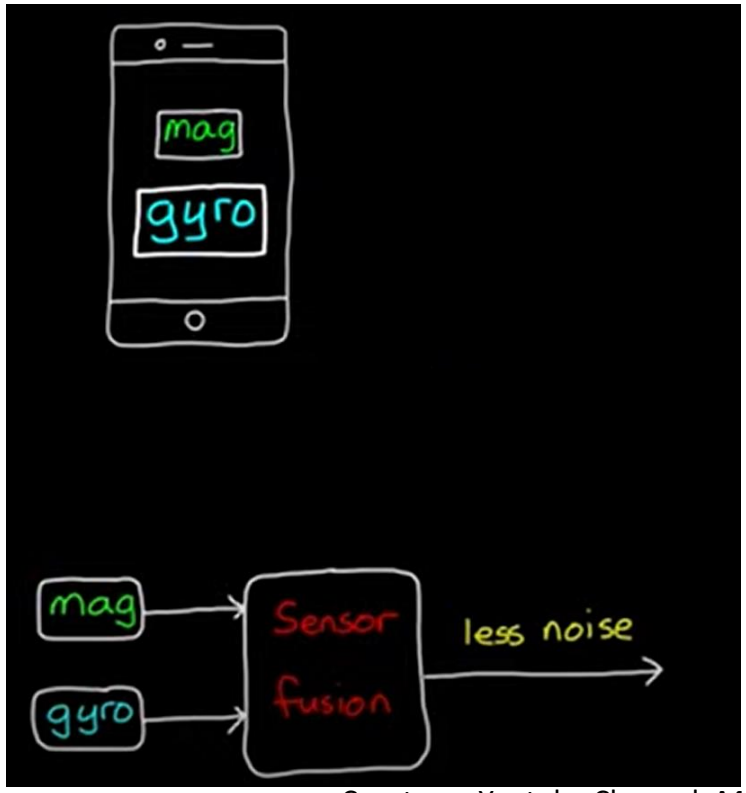

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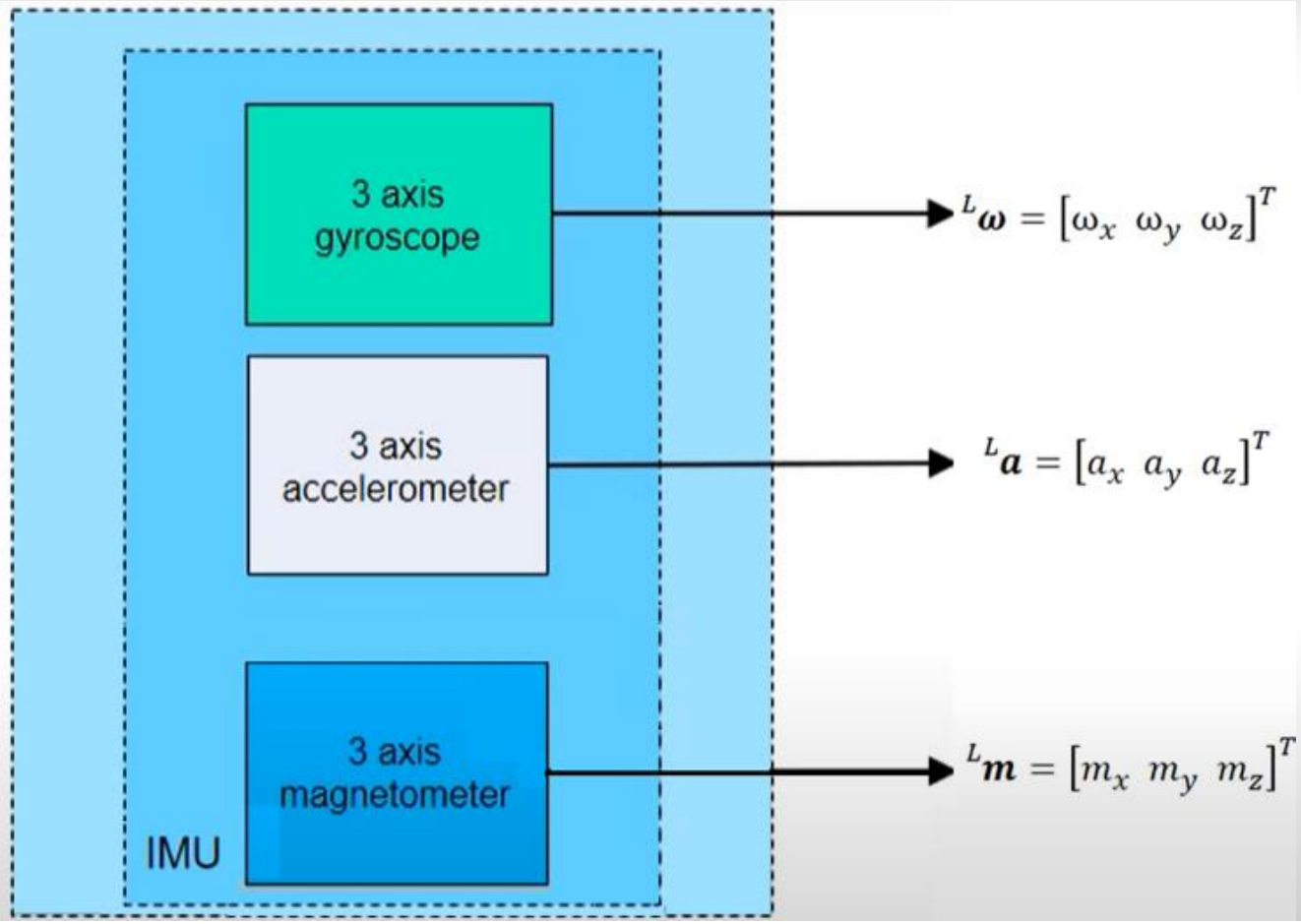


$$\begin{cases} {}^{L}\boldsymbol{\omega} = {}^{L}\boldsymbol{\omega}_{true} + \boldsymbol{b}_g + \boldsymbol{v}_g \\ {}^{L}\boldsymbol{a} = {}^{L}\boldsymbol{a}_{body} + {}^{L}\boldsymbol{g} + \boldsymbol{b}_a + \boldsymbol{v}_a \\ {}^{L}\boldsymbol{m} = {}^{L}\boldsymbol{m}_{true} + {}^{L}\boldsymbol{m}_{int} + \boldsymbol{b}_m + \boldsymbol{v}_m \end{cases}$$

${}^{L}\boldsymbol{\omega}$ = Angular velocity vector in the local (sensor) frame

${}^{L}\boldsymbol{a}$ = Acceleration vector in the local frame

${}^{L}\boldsymbol{m}$ = Magnetic field vector in the local frame

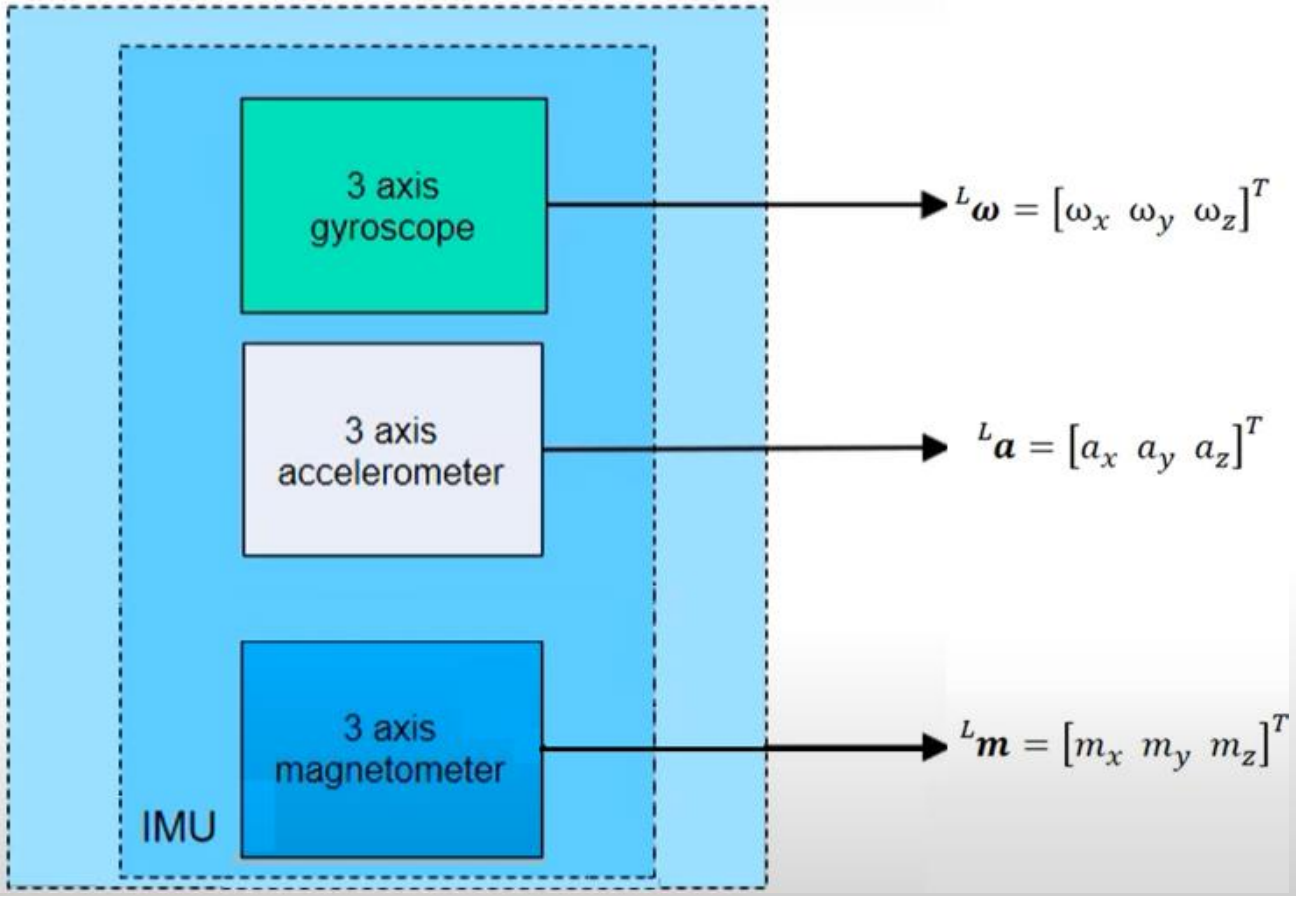
$\boldsymbol{b}$ = sensor bias

$\boldsymbol{v}$ = sensor noise

Courtesy : Mathworks: Youtube Academic Video: Sensor Fusion for Orientation Estimation

• Sensor Fusion (Attitude Heading and Reference System (AHRS)

Micro-electro-mechanical system(MEMS) based device



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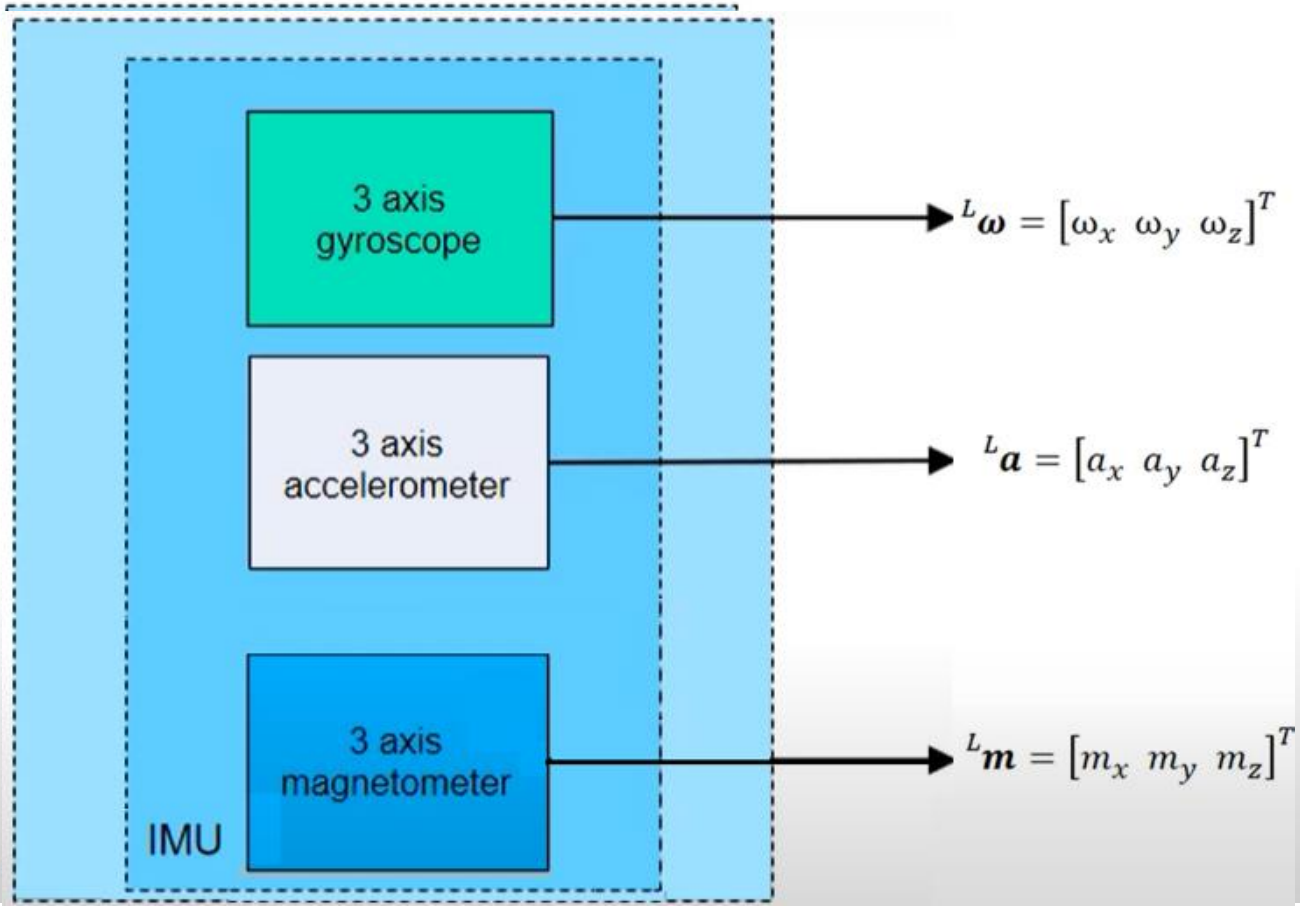
• Sensor Fusion (Attitude Heading and Reference System (AHRS)

Micro-electro-mechanical system(MEMS) based device



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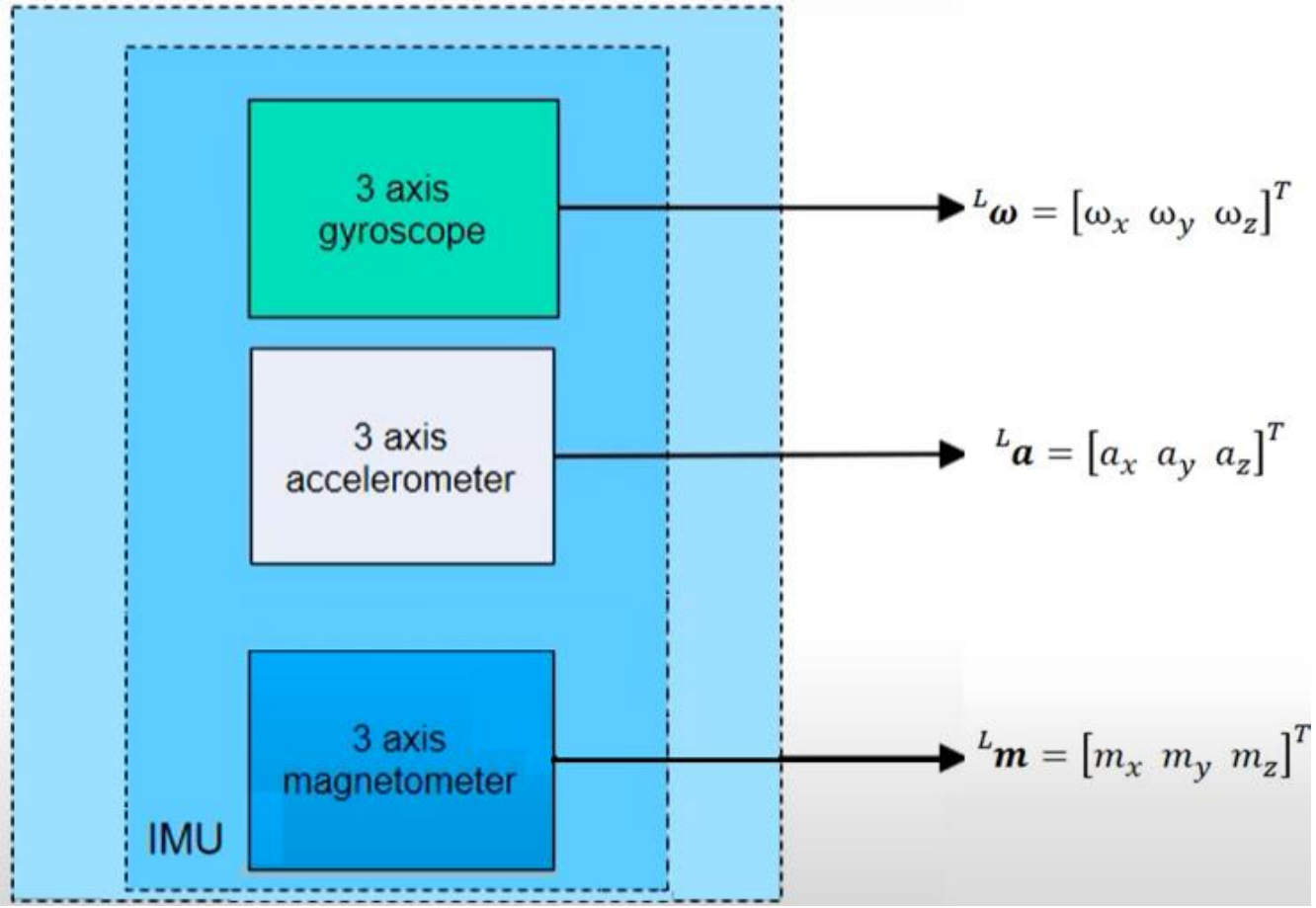
• Sensor Fusion (Attitude Heading and Reference System (AHRS)

Micro-electro-mechanical system(MEMS) based device



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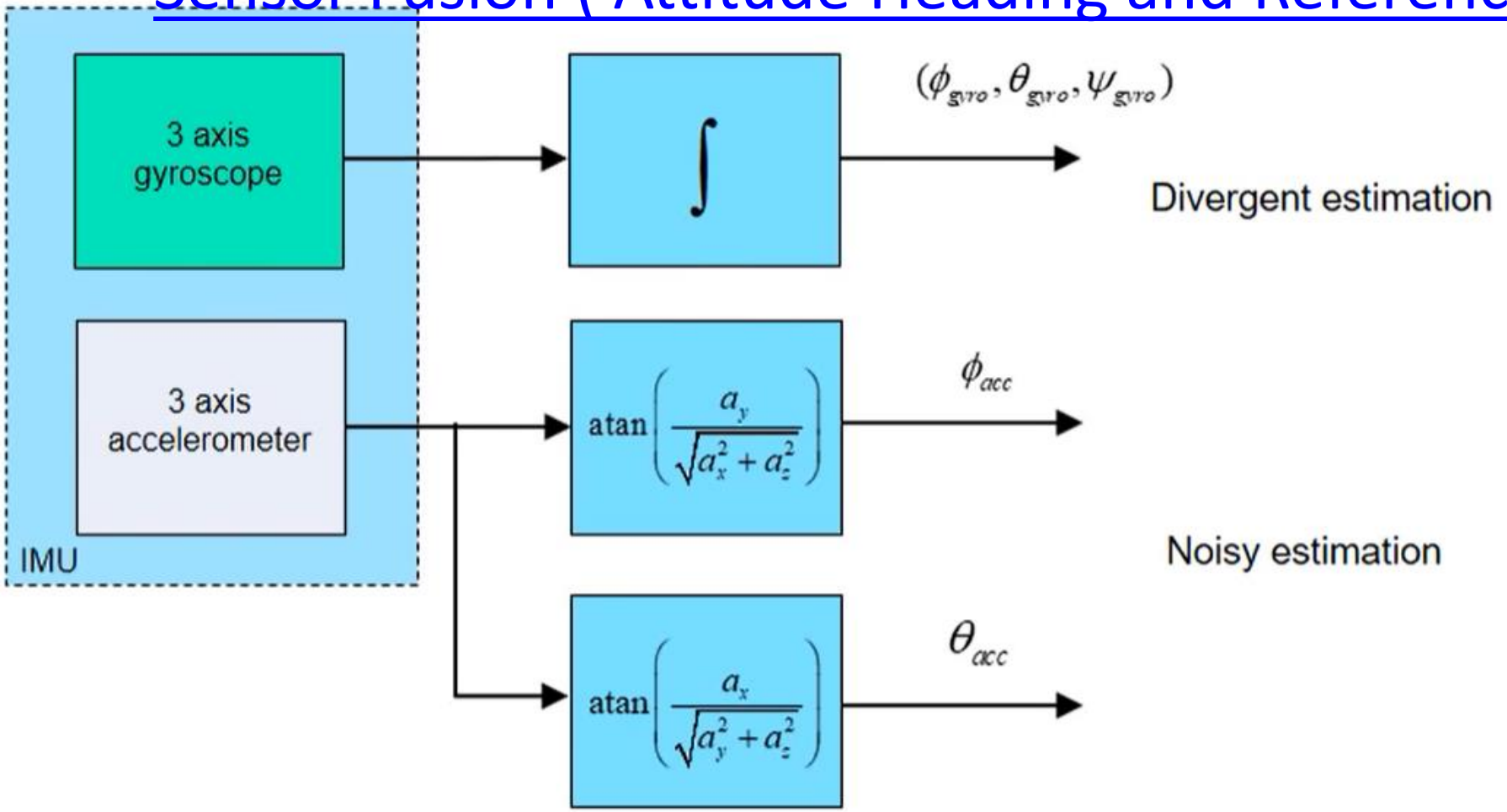


Euler Angles

ϕ roll

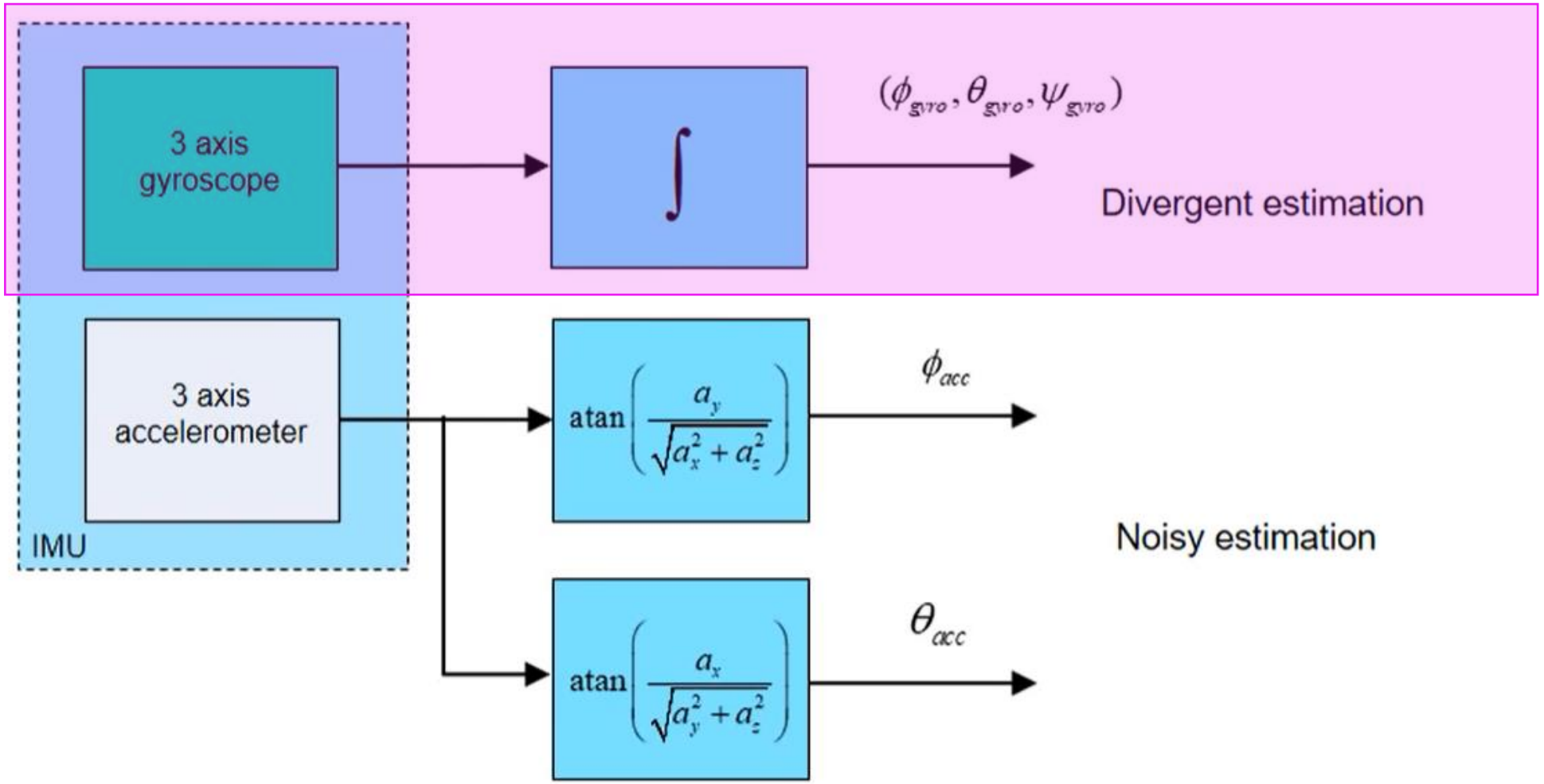
θ pitch

ψ yaw

Assumption: Total acceleration = gravity (measured in the body frame)

Courtesy : Mathworks: Youtube Academic Video: Sensor Fusion for Orientation Estimation

• Sensor Fusion (Attitude Heading and Reference System (AHRS)



Euler Angles

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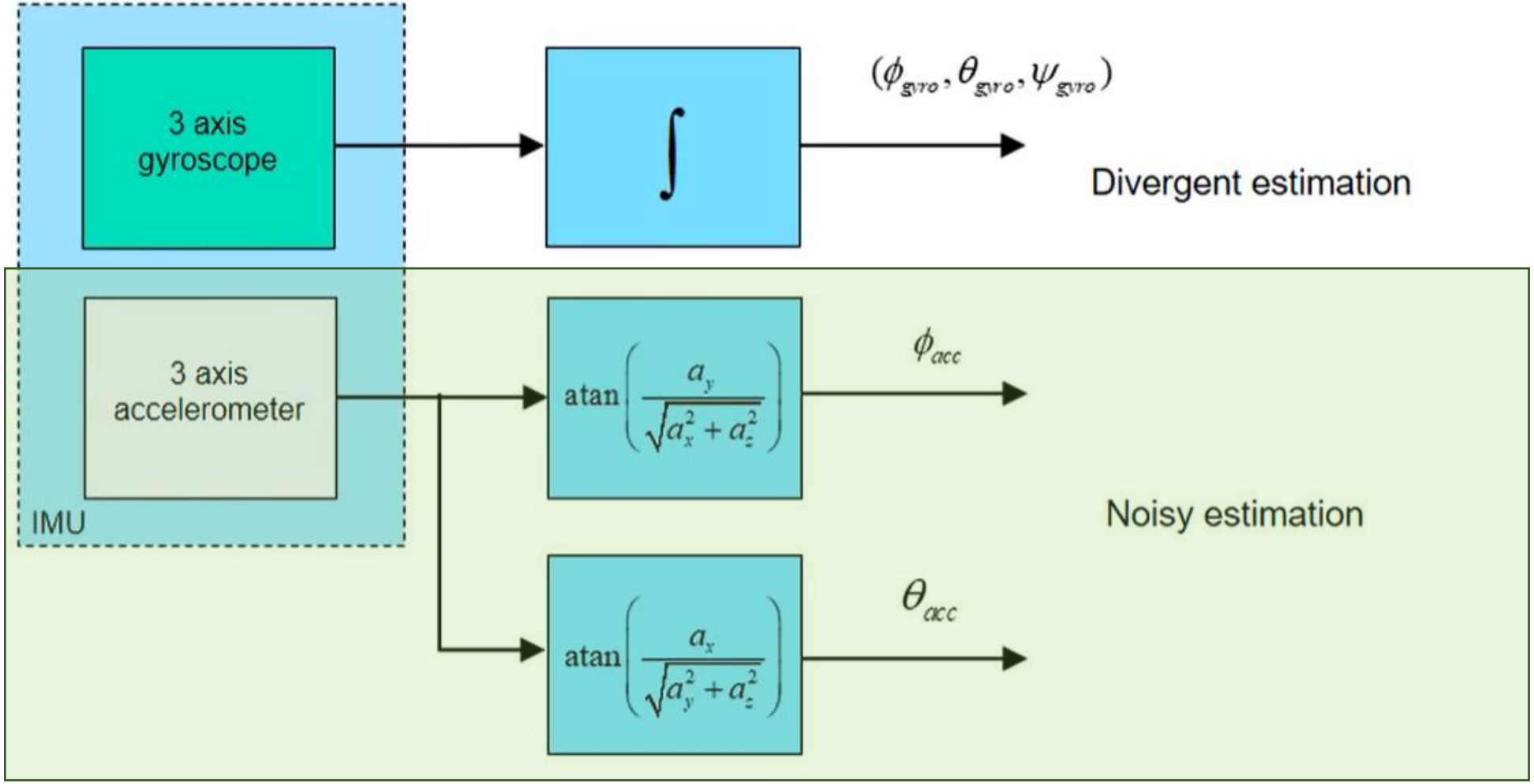
Courtesy : Mathworks: Youtube Academic Video: Sensor Fusion for Orientation Estimation

Assumption: Total acceleration = gravity (measured in the body frame)

• Sensor Fusion (Attitude Heading and Reference System (AHRS)



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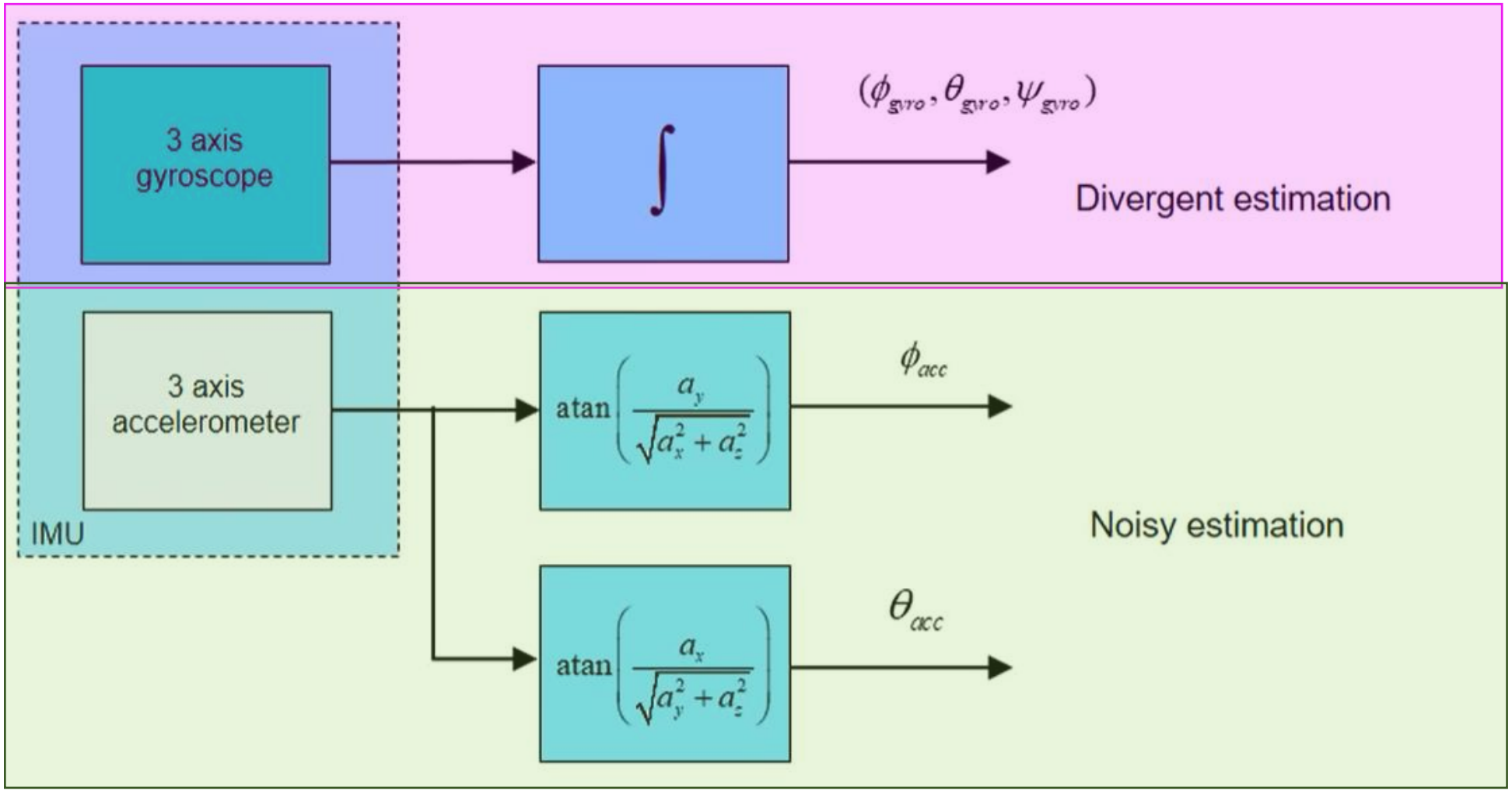

Euler Angles

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Assumption: Total acceleration = gravity (measured in the body frame)

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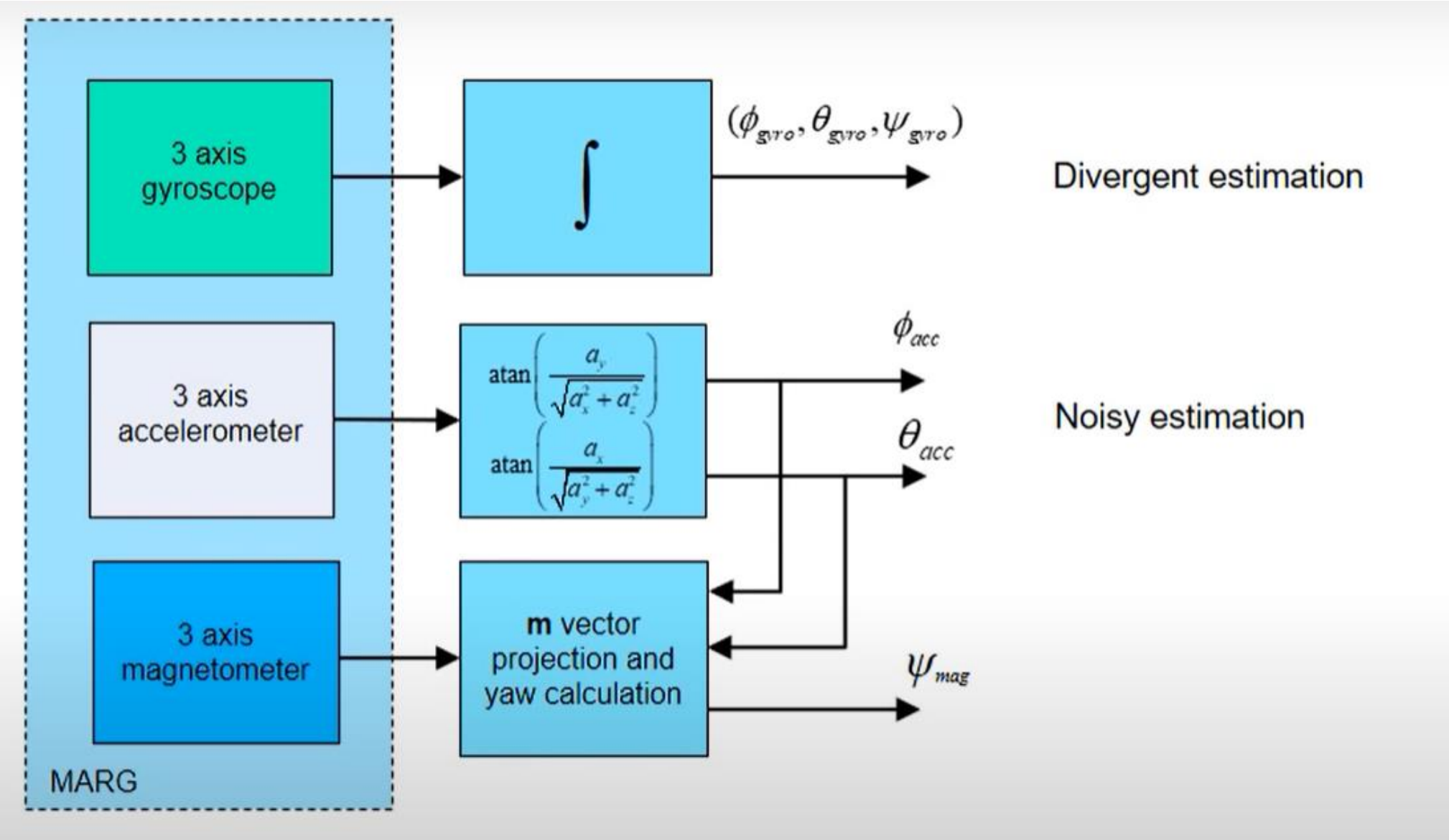
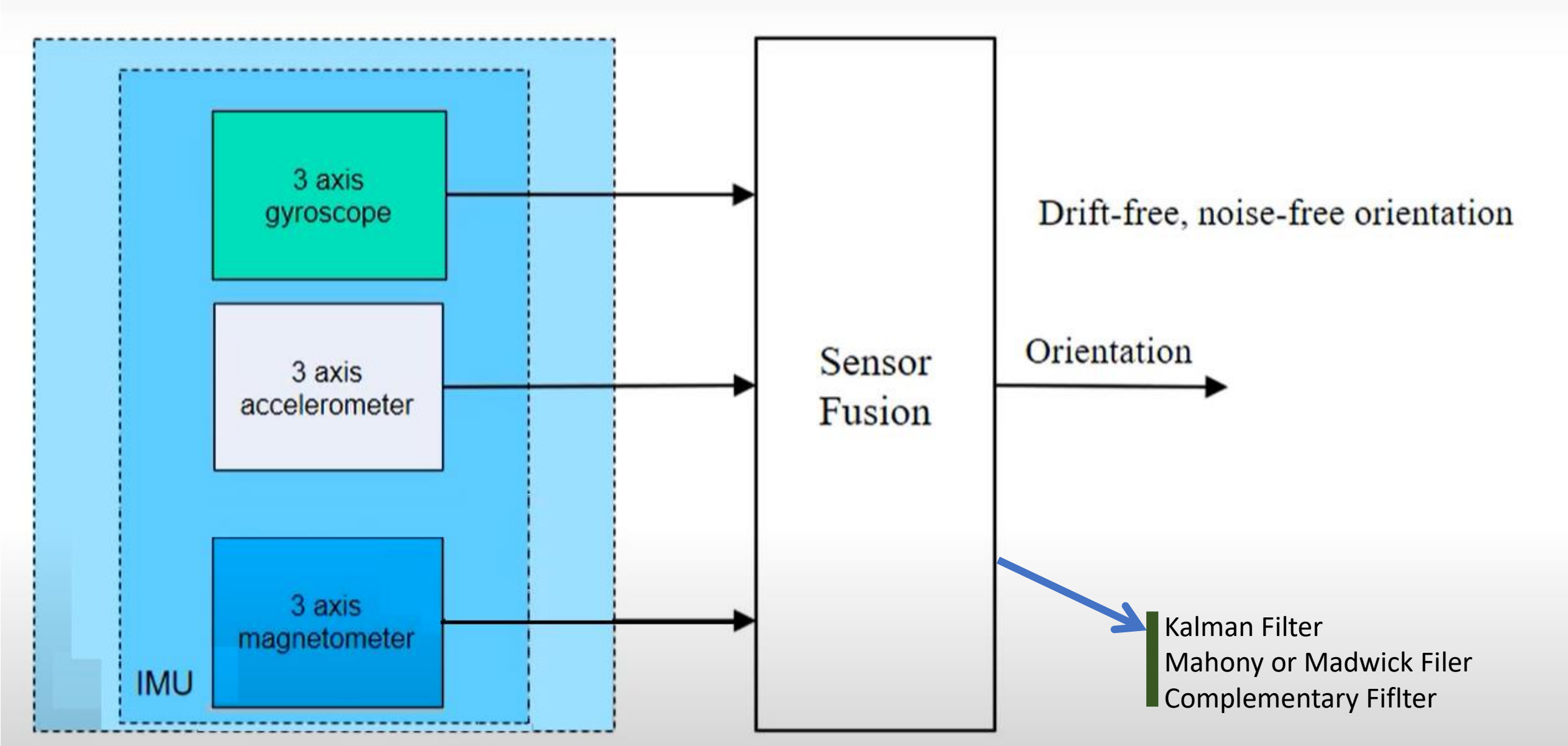
• Sensor Fusion (Attitude Heading and Reference System (AHRS)



Courtesy : Mathworks: Youtube Academic Video: Sensor Fusion for Orientation Estimation

• Sensor Fusion (Attitude Heading and Reference System (AHRS)

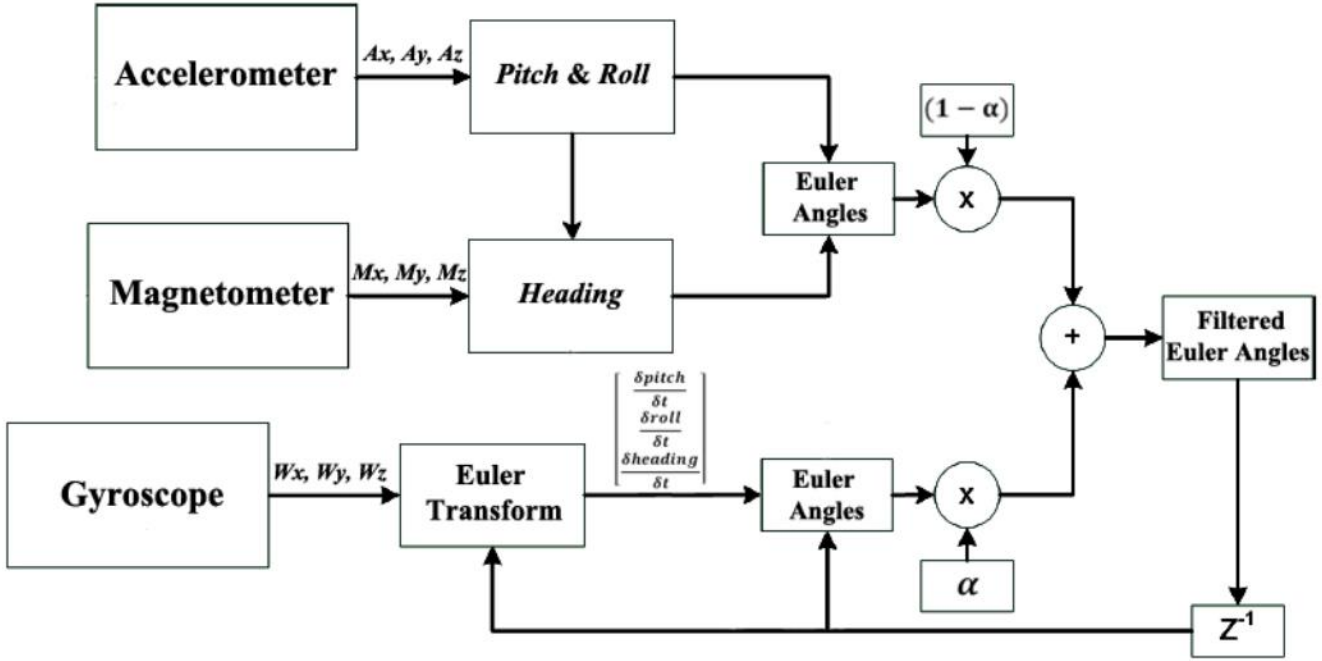


Courtesy : Mathworks: Youtube Academic Video: Sensor Fusion for Orientation Estimation

• Sensor Fusion (Attitude Heading and Reference System (AHRS)

• Data fusion with complementary filter uses two static gain components sum to one to merge orientation achieved from gyroscope measurements with the one provided by accelerometer and magnetometer as following:

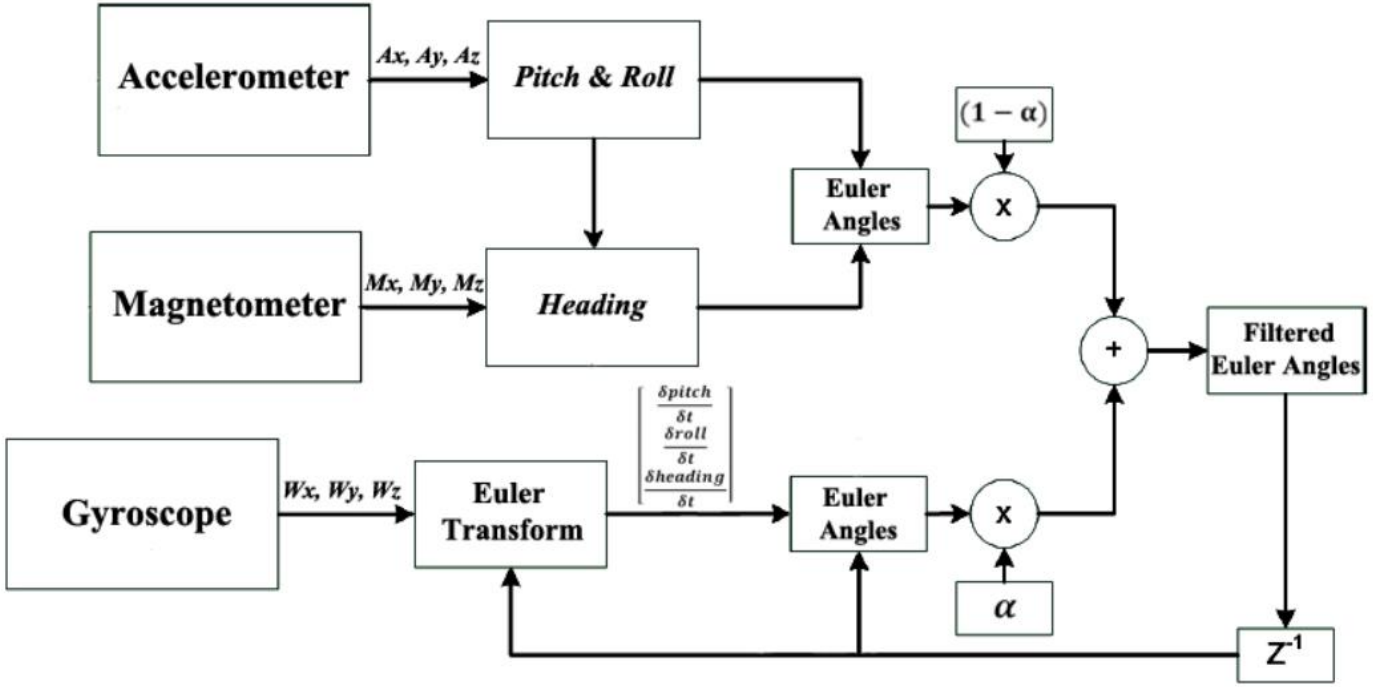
$$\begin{bmatrix} pitch \\ roll \\ heading \end{bmatrix}_{Filtered} = \alpha . \begin{bmatrix} pitch \\ roll \\ heading \end{bmatrix}_{Gyro} + (1-\alpha) . \begin{bmatrix} pitch \\ roll \\ heading \end{bmatrix}_{Acc\ \&\ Mag}$$



• Sensor Fusion (Attitude Heading and Reference System (AHRS)

● **Filtering Algorithm Consideration:**

* The data need to be fused should be expressed in the same convention.

* Fast changes related to gyroscope are more reliable than those achieved by accelerometer and magnetometer. High static gain of $0.9 < \alpha < 1.0$ allows the system to respond against fast changes in measurements.

* In order to avoid divergences due to gyroscope bias, steady convergence must be achieved using accelerometer and magnetometer measurements rather than gyroscope ones. This can be achieved by feedback the filtered angles to be used in the gyroscope measurements.